AdventureWorksLT – SQL Practice Assignment

15 Questions | SQL Server | Easy, Intermediate & Advanced

Database	AdventureWorksLT
Total questions	15
Difficulty split	5 Easy • 8 Intermediate • 2 Advanced
Instructions	Write and execute a SQL query for every question. Use the tables available in your AdventureWorksLT database.

Tables available in the database

SalesLT.Address	SalesLT.Customer
SalesLT.CustomerAddress	SalesLT.Product
SalesLT.ProductCategory	SalesLT.ProductDescription
SalesLT.ProductModel	SalesLT.ProductModelProductDescription
SalesLT.SalesOrderDetail	SalesLT.SalesOrderHeader

Note: Use the SalesLT schema prefix when referencing tables. Column names may vary slightly if your database is a customized AdventureWorksLT version.

Section A – Easy (5 Questions)

Question 1. Display all columns for products whose ListPrice is greater than 1,000. Sort the results by ListPrice from highest to lowest.

Question 2. Display the CustomerID, FirstName, LastName, and CompanyName of customers whose LastName starts with the letter 'S'.

Question 3. List all sales orders placed on or after 1 January 2008. Show SalesOrderID, CustomerID, OrderDate, and TotalDue, sorted by OrderDate.

Question 4. Display each distinct product Color available in the Product table. Exclude rows where Color is NULL.

Question 5. For each product, calculate the difference between ListPrice and StandardCost. Display ProductID, Name, ListPrice, StandardCost, and the calculated difference. Show only products where ListPrice is greater than StandardCost.

Section B – Intermediate (8 Questions)

Question 6. Show the total number of sales orders placed by each customer. Display CustomerID and the order count. Include only customers who have placed at least two orders.

Question 7. Find the total quantity sold for each product using SalesOrderDetail. Display ProductID and total quantity sold. Show only products with a total quantity greater than 50.

Question 8. Display the number of products in each ProductCategory. Include the category name and product count. Include categories that have at least one product.

Question 9. Find customers whose total order value (sum of SalesOrderHeader.TotalDue) is greater than 10,000. Display CustomerID and total order value.

Question 10. Using a subquery, display products whose ListPrice is greater than the average ListPrice of all products.

Question 11. Using a CTE, calculate the total sales amount for each customer from SalesOrderHeader. Return only customers whose total sales amount is above the overall average customer sales amount.

Question 12. Rank products within each ProductCategory based on ListPrice from highest to lowest. Display ProductID, Name, ProductCategoryID, ListPrice, and the rank. Use RANK().

Question 13. For each customer, display their orders in date order and show the previous order date alongside the current order date. Use the appropriate window function and display CustomerID, SalesOrderID, OrderDate, and previous order date.

Section C – Advanced (2 Questions)

Question 14. For each customer, show every order along with its order value and the customer's running total of order value in chronological order. Use SUM() OVER with PARTITION BY and an appropriate window ORDER BY. Include CustomerID, SalesOrderID, OrderDate, TotalDue, and running total.

Question 15. Identify the top three products by total sales revenue within each ProductCategory. Calculate revenue from SalesOrderDetail using OrderQty, UnitPrice, and UnitPriceDiscount. Use a CTE or subquery together with a ranking window function. Return category, product, revenue, and rank.

Submission checklist

■ Submit one SQL query for each question.

■ Use meaningful column aliases where appropriate.

■ Ensure the query executes successfully in SQL Server.

■ Do not modify the source tables or data.